AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 3 · TEACHER GUIDE

Branching Logic

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Teach branch order with visible assignments and skipped checks.

Provide executable correct/faulty scripts and distinguish each example's boundary before independent transfer.

Keep one habit: Look for the first true branch, correct indentation and a revealing equality test.

MAIN LESSON · SECTION 1

Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

Learning goal and precision

Primary target: Trace the selected **branch** in an if elif else chain

Teach **order**ed selection and **indentation** using a written **input** contract. A script can cover all **branch**es while missing an equality bug. Reference: https://docs.python.org/3/tutorial/controlflow.html#if-statements.

The new task requires one four-way chain; independently correct equivalent **condition**s are acceptable.

By the end, I can **order branch**es so each **input** receives exactly its intended category.

What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.
- Case exposure check: this source example appears in ai-2/3, python-bridge/5. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.
- Python 3 editor and **branch**es.py
- The written duration rule and **boundary** case cards
- Blank trace table with a skipped marker

Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Opening retrieval and readiness

Without opening the old lesson, explain one key idea from Decompose a Goal. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish a function definition from a function call

Reference example, not the only acceptable wording: Name show_remaining; parameters capacity and used. Arguments 15 and 6 match capacity = 15 and used = 6. Expected printed result 9.

<= accepts equality; < excludes it. At 12 the two **condition**s produce different Boolean values.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Decompose a Goal

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Classify lengths with if pages<80 then Short; else if pages<=160 then Medium; else Long. Trace 79,80,160,161. What happens if the <=160 **branch** is placed first?

Concept to explain

Ordered **condition**s determine which path a system takes.

Term	Meaning
condition	A statement evaluated as true or false to help select an action.
branch	One path taken when a condition selects among alternatives.
if/else	A programming structure that chooses a path according to whether a condition is true or false.
order	The sequence in which steps are performed or items are read.

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach branch order through the complete script and exact boundaries

> Keep the main twelve-minute rule, separate ten-minute rule and independent tile rule labelled.

State the main rule and Python syntax

A fictional activity sorter labels a supplied integer duration. Negative values are Invalid; values from zero through 12 are Short; larger values are Long. The labels describe duration only.

Write this requirement before the program so that a test has an expected answer independent of the code.

An if statement selects a block when its **condition** is true. An elif is considered only if earlier **condition**s in the same chain were false. An else handles the remaining cases. Colons end the **condition** lines.

Use four spaces for the indented statements and keep related **branch**es aligned.

Read the complete executable script

```
minutes = 12
if minutes < 0:
    label = "Invalid"
elif minutes <= 12:
    label = "Short"
```

```
else:
    label = "Long"
print(label)
```

Trace selected and skipped paths

The **output** is Short. For 12, the first **condition** is false and the second is true. The else block is skipped. print(label) is not indented, so it runs after the selected **branch**.

Each path assigns label before this line reads it. Change only minutes and save before each fresh run.

Input minutes	Expected label	Reason for this test
-1	Invalid	Negative input rejection
0	Short	Lowest valid duration
11	Short	Just below the upper boundary
12	Short	Exact upper boundary
13	Long	Just above the boundary

These tests exercise every **branch** and the equality case. A change from <= 12 to < 12 is exposed by 12; testing 11 alone misses it.

Branch coverage means every **branch** was exercised, but it does not prove every **condition** is correct.

> **Suppose the program uses < 12 instead of <= 12. What is printed for 12, why does that violate the written rule, and which earlier test should still pass after repair?**

AI ACADEMY / AI-2/3 · 03 · INSPECT THE LESSON MODEL

One chain chooses one label

Input	Checks in order	Selected label
-1	Below zero is True	Invalid
12	Below zero False; <= 12 True	Short
13	Both checks False	Long

Later branches in this chain are skipped after a condition succeeds.

Follow the labelled picture with the matching explanation. Open the vector diagram

Introduce the separate reading-time rule

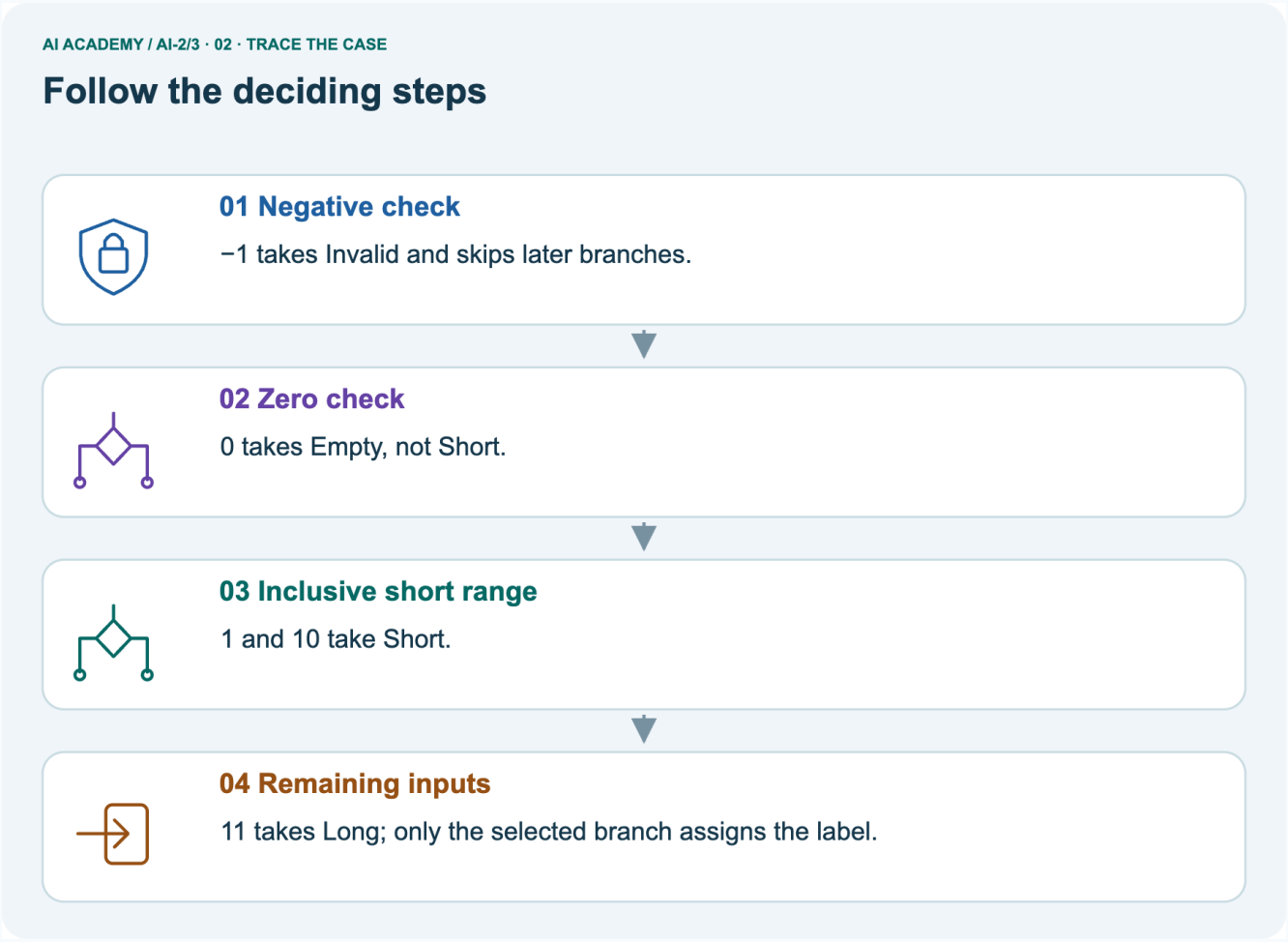
A fictional reading-time labeler uses one Python if/el**if/else** chain.

Whole minutes below 0: Invalid; 0: Empty; 1–10: Short; above 10: Long.

Order checks as minutes<0, minutes==0, minutes<=10, otherwise. Print the assigned label once.

```
minutes = 0
if minutes < 0:
    label = "Invalid"
elif minutes == 0:
    label = "Empty"
elif minutes <= 10:
    label = "Short"
else:
    label = "Long"
print(label)
```

Trace branch precedence and category coverage



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Negative check	–1 takes Invalid and skips later branches.
2. Zero check	0 takes Empty, not Short.
3. Inclusive short range	1 and 10 take Short.
4. Remaining inputs	11 takes Long; only the selected branch assigns the label.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

The third comparison alone would include negative values and zero, but earlier **branch**es already handle them. An elif chain selects its first true **branch**.

Multiple independent if statements would have different behavior and could overwrite an earlier label. **Boundary** traces show why **order** belongs to the specification.

Keep the input-type limit explicit

The example assumes whole numeric minutes; strings, missing values and fractional values need a separate declared **input** contract.

Move the broad condition first

Change: Move minutes<=10 to the first **branch**.

Prediction: –1 and 0 wrongly receive Short.

Reason: Both satisfy the earlier comparison before their intended checks are reached.

Use pictures to elicit deciding evidence

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied **condition** and compare.

The picture is a representation; it does not document execution of a real AI model.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Explain the mechanism

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. IF pet is hungry → feed.

2. ELSE IF pet needs exercise → walk or play.

3. ELSE → provide a quiet rest space.

4. **Boundary** test: what if hungry and needs exercise? The first true **branch** wins.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

If pages<100 choose Short; otherwise if pages<=200 choose Medium; otherwise Long.

Classify 99,100,200,201 pages.

Exact explanation to model

Short, Medium, Medium, Long. Strict and inclusive boundaries and **branch order** matter.
</details>

Explain the added worked example

Key terms

Term	Meaning
Branch	A possible path through a program
Condition	An expression used to decide which path to take
Indentation	Leading spaces that identify a Python code block
Boundary	A value where a classification may change
Regression check	A previous test rerun after an edit to detect a new failure

Model the reasoning aloud

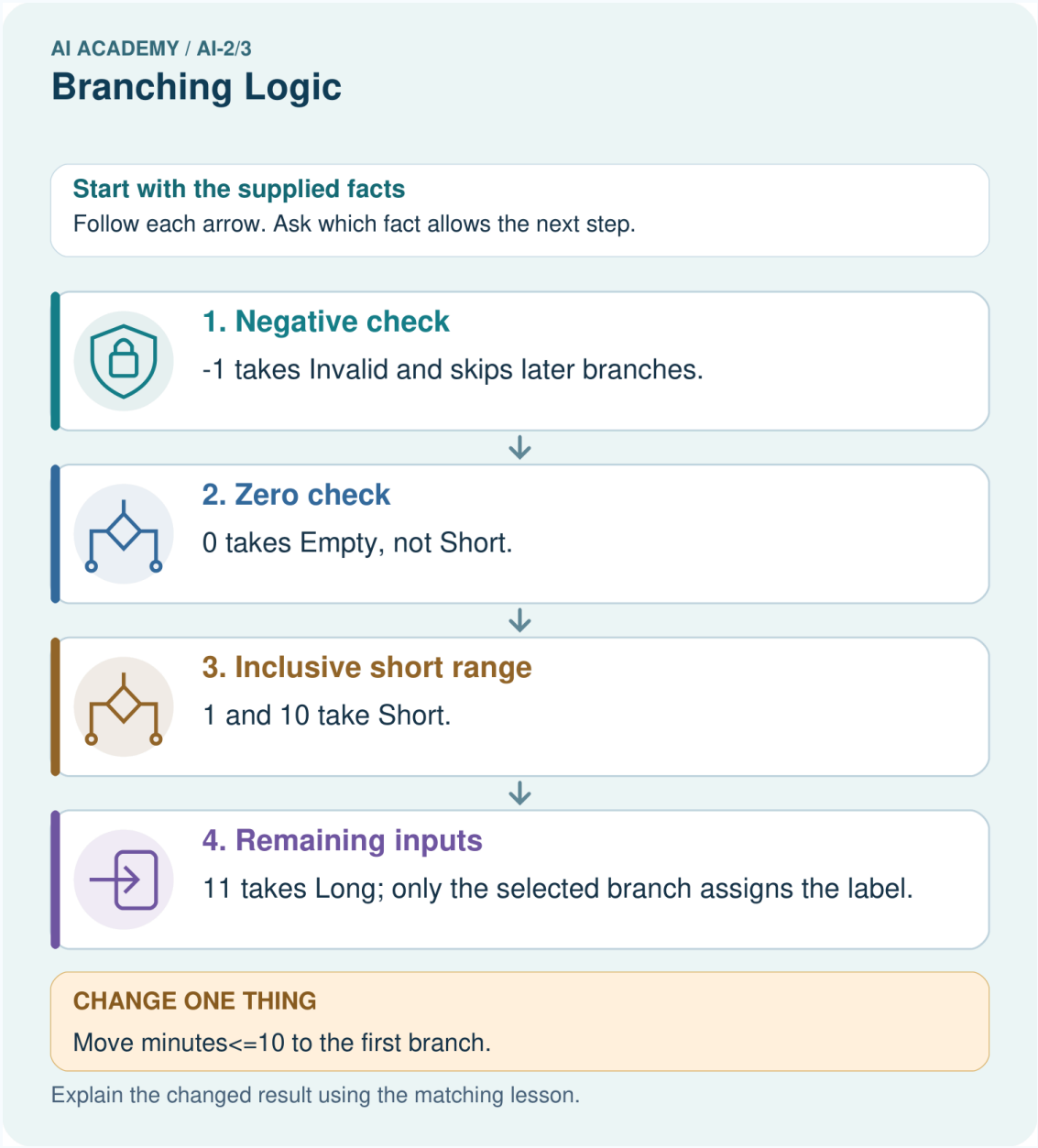
Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

A pictorial worked case: Branching Logic

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

MAIN LESSON · SECTION 3

Guide the reasoning, then remove help

> Collect a response from each learner before discussing the worked key.

Prompt before revealing

Completion task: For -1, 12 and 13, record checked **condition**s, skipped **condition**s and the final label. Run each as a fresh script after predicting.

Expected reasoning and fading help

Teacher check: -1: first True, second skipped, Invalid. 12: first False, second True, Short. 13: both False, else selected, Long. Retain observed results separately.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Check the misconception before discussion

Use the worked case for **Branch**ing and **boundary** tests. Proposed explanation: “Every **condition** in the chain always runs”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Once one **condition** succeeds, later alternatives are skipped.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Use the pictorial check

Give expected labels for –1,0,10,11 and identify the equality **boundary** for Short.

Invalid, Empty, Short, Long. Ten is included in Short because the comparison uses <=.

Changed-case reasoning: –1 and 0 wrongly receive Short. Both satisfy the earlier comparison before their intended checks are reached.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

- Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.
- Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.
- In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Demonstrate and guide practice

Board sequence

Write the given facts: If pages<100 choose Short; otherwise if pages<=200 choose Medium; otherwise Long.

Write the question to explain: Classify 99,100,200,201 pages.

Reveal after **prediction**s: Short, Medium, Medium, Long. Strict and inclusive boundaries and **branch order** matter.

Use the earlier changed question as optional later practice. Reserve Build a new sorter independently for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace **condition**s for decisions, and identify supporting sources for claims.

Model these guided questions

The faulty < 12 version prints Long for 12, contrary to the inclusive Short rule. Repair to <= 12 and rerun 11 or 13 as well as 12.

Exit: exact 12 is missing. -1, 2 and 20 can visit all **branch**es even when the equality comparison is wrong.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve inclusive comparison and collect the prerequisite.
7 to 20 minutes	Translate the duration rule and complete W1.
20 to 33 minutes	Trace the diagram and code, marking skipped branches.
33 to 45 minutes	Complete W2 with predicted and actual outputs.
Session 2 start to 8 minutes	Retrieve why equality needs a separate test.
8 to 20 minutes	Run the faulty two-if script and complete W3 and the guided question.
20 to 35 minutes	Collect the new sorter in W4 before hints.
35 to 45 minutes	Use W5 and the exit to distinguish coverage from correctness.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied investigation

> Keep expected results, observed paper traces and remaining limits distinguishable.

Supplied materials and answer guidance

For the fictional pet-token cards, hungry-first gives Feed,Feed,Play,Rest,Feed,Play. Play-first changes A and E to Play; other tokens stay the same. These are paper algorithm outcomes, not animal-care advice. Main duration outputs for -1,0,11,12,13 are Invalid,Short,Short,Short,Long. The two-if version overwrites Invalid with Short at -1; restore elif and retain the failed trace. A strict <12 version fails exactly 12. Actual runs must be recorded rather than copied from expectations. Reserve Build a new sorter independently for the tile-category assessment.

Run and record

1. Predict → Test → Record → Explain
2. Keep expected and actual results separate. Surprises are useful data when recorded honestly.
3. Trace six pet cards through the same **order**ed rules. Change only the rule **order** and compare **output**s.

4. What did the hardest test teach you that the normal test did not?

Keep an inspectable record

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the first misunderstanding

Every **condition** in the chain always runs: Once one **condition** succeeds, later alternatives are skipped.

Two if statements behave like if then elif: Independent if statements can both run and overwrite a result.

Indentation is cosmetic: Leading spaces define the **branch** body.

Every **branch** tested means every **input** correct: A missing equality case can hide a wrong comparison.

Choose support from the response

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a **boundary** or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Build a new sorter independently for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected new case

> Keep this key until after the first attempt; accept valid alternatives with defensible reasoning.

Selected independent question and key

Selected case: Use an integer tiles count. Label a negative value Invalid, zero Empty, 1 through 6 Small and values above 6 Large. Write an **order**ed if elif else script and predict -1, 0, 1, 6 and 7.

Include one print after the chain.

Use if tiles < 0, elif tiles == 0, elif tiles <= 6, else, assigning the respective labels and printing after the chain. **Output**s Invalid, Empty, Small, Small, Large. Zero must be handled before the <= 6 **branch**.

Judge explanation, trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.

Dimension	Evidence required
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the chain

Describe the purpose of if, elif and else in the worked duration sorter. Explain why print(label) is outside the indented **branch**es.

if tests negative first; elif checks <= 12 only after that fails; else handles the rest. Unindented print runs after any selected **branch**, whose assignment gives label a value.

W2 Trace the tested path

For -1, 12 and 13, record checked **condition**s, skipped **condition**s and the final label. Run each as a fresh script after predicting.

-1: first True, second skipped, Invalid. 12: first False, second True, Short. 13: both False, else selected, Long. Retain observed results separately.

W3 Repair the overwritten result

Trace the two-independent-if example for -1. Identify which assignment overwrites the earlier label and make the smallest repair.

The first assignment sets Invalid; the second if also succeeds and changes label to Short. Replace the second if with elif. The repaired **output** is Invalid.

W5 Choose revealing regression tests

A repair changes the small-count **condition** from < 6 to <= 6. Name the case that exposes the original error and two different kinds of earlier cases to rerun.

6 exposes the original wrong Large label. Rerun zero for Empty and a negative count for Invalid, or 7 for Large. Passing only 6 does not check earlier **branch**es.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

Short, Medium, Medium, Long. Strict and inclusive boundaries and **branch order** matter.

Additional transfer practice question

If pages are at most 80, choose short; otherwise choose long. Classify books with 79, 80 and 81 pages and explain the **boundary**.

Reasoned practice answer

79 and 80 are short; 81 is long. At most 80 includes 80.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “At most 80 excludes 80.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Original results are Short, Medium, Medium, Long. With <=160 first, 79 is incorrectly routed to Medium and the Short **branch** becomes unreachable for nonnegative short books. Check values just below, at and above each **boundary**.

Explanations for the additional workbook examples

W1 Explain the chain

W2 Trace the tested path

W3 Repair the overwritten result

W4 Build a new sorter independently

Use an integer tiles count. Label a negative value Invalid, zero Empty, 1 through 6 Small and values above 6 Large. Write an **order**ed if elif else script and predict -1, 0, 1, 6 and 7. Include one print after the chain.

Use if tiles < 0, elif tiles == 0, elif tiles <= 6, else, assigning the respective labels and printing after the chain. **Output**s Invalid, Empty, Small, Small, Large. Zero must be handled before the <= 6 **branch**.

W5 Choose revealing regression tests

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

Project evidence to keep

Branch when a reader has a hard constraint such as language, format, or maximum length.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Validate recommender requests before ranking items, retaining explicit labels for missing or invalid cases.

Support while preserving the reasoning goal

Trace one record or **condition** at a time with counters and labelled columns. Stretch: introduce a tie, **boundary** or missing field and declare its handling policy before rerunning.

Provide movable **branch** cards and align the four-space blocks on a ruled page. Permit an oral trace alongside the script and record help.

Extension: swap the zero and <= 6 checks and find the smallest nonnegative counterexample.

W4 must assign one correct label on all five cases and explain zero before the small **branch**. After support, retry a shelf-count sorter with Empty at zero, Few for 1 through 4, Many above 4 and Invalid below zero.

Expected **output**s at -1, 0, 4, 5 are Invalid, Empty, Few, Many.

Plan the delayed check

Suggested return points in this level: ai-2/4, ai-2/6, ai-2/9. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 4: Trace Tables. Retrieve today's **evidence** and identify what carries forward and what is new.

Weeks 4 and 5 trace repeated blocks, loop state and stopping.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Transparent Book Recommender

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

Preparation and schedule

Teaching target: Trace the selected **branch** in an if elif else chain

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

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> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
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>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Practice and assessment reminders

Before class, solve the supported practice: For -1, 12 and 13, record checked **condition**s, skipped **condition**s and the final label. Run each as a fresh script after predicting.

Reasoned practice answer: -1: first True, second skipped, Invalid. 12: first False, second True, Short. 13: both False, else selected, Long. Retain observed results separately.

Check every learner before discussion: Use the worked case for **Branch**ing and **boundary** tests. Proposed explanation: “Every **condition** in the chain always runs”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Once one **condition** succeeds, later alternatives are skipped.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Move minutes<=10 to the first **branch**. Require a **prediction** and a reason before comparing. Expected result: –1 and 0 wrongly receive Short.

Reason: Both satisfy the earlier comparison before their intended checks are reached.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference: feedback and records**

REFERENCE · SECTION 8

Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The **branch-boundary** and indexing lessons deliberately include failure cases.
</details>

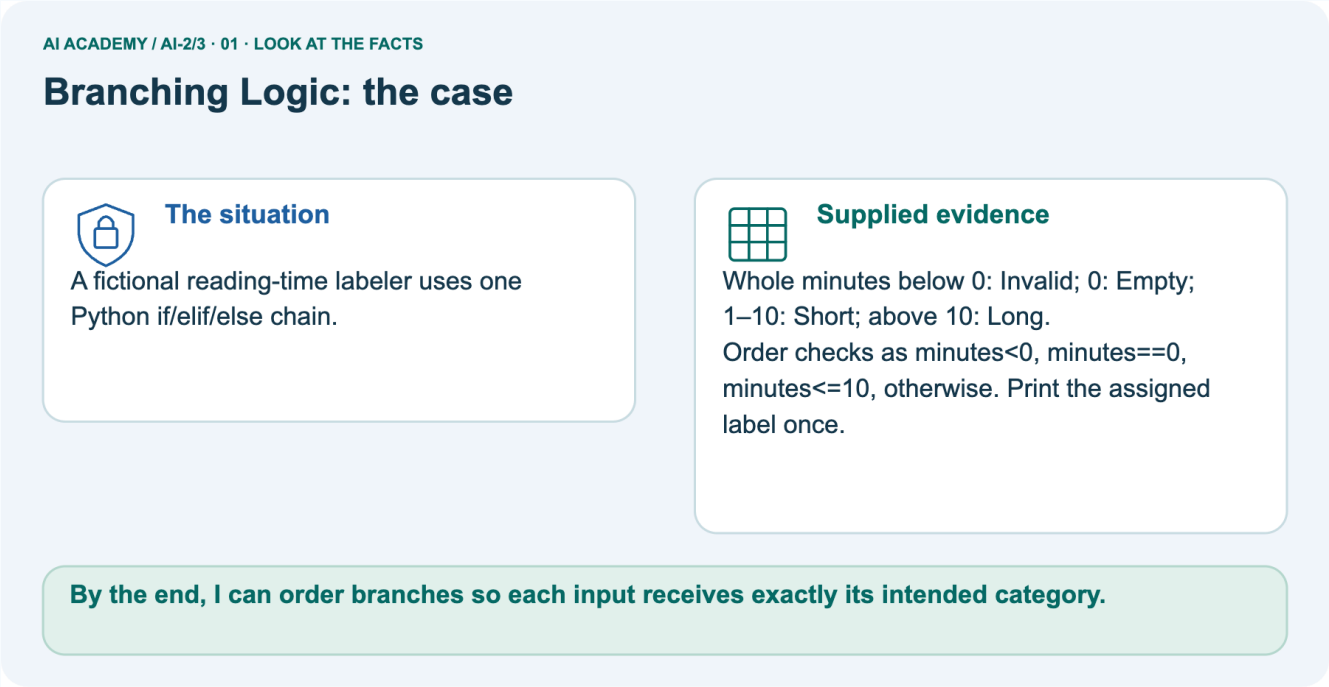
<details>
<summary>Reference: teach with the six pictures</summary>
</details>

REFERENCE · SECTION 9

Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

01. Branching Logic: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

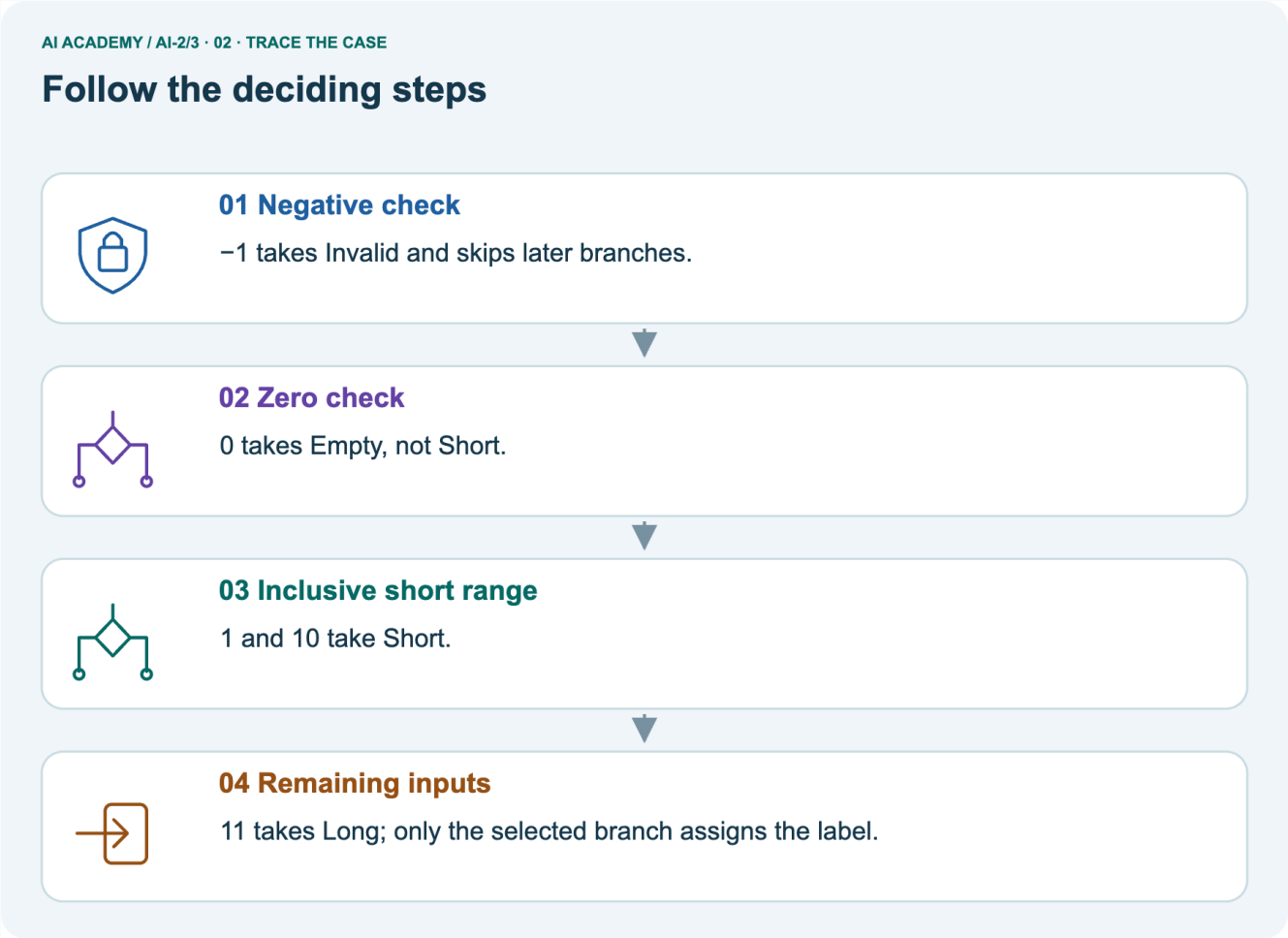
A fictional reading-time labeler uses one Python if/el**if/else** chain. Supplied facts: Whole minutes below 0: Invalid; 0: Empty; 1–10: Short; above 10: Long. **Order** checks as minutes<0, minutes==0, minutes<=10, otherwise.

Print the assigned label once.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



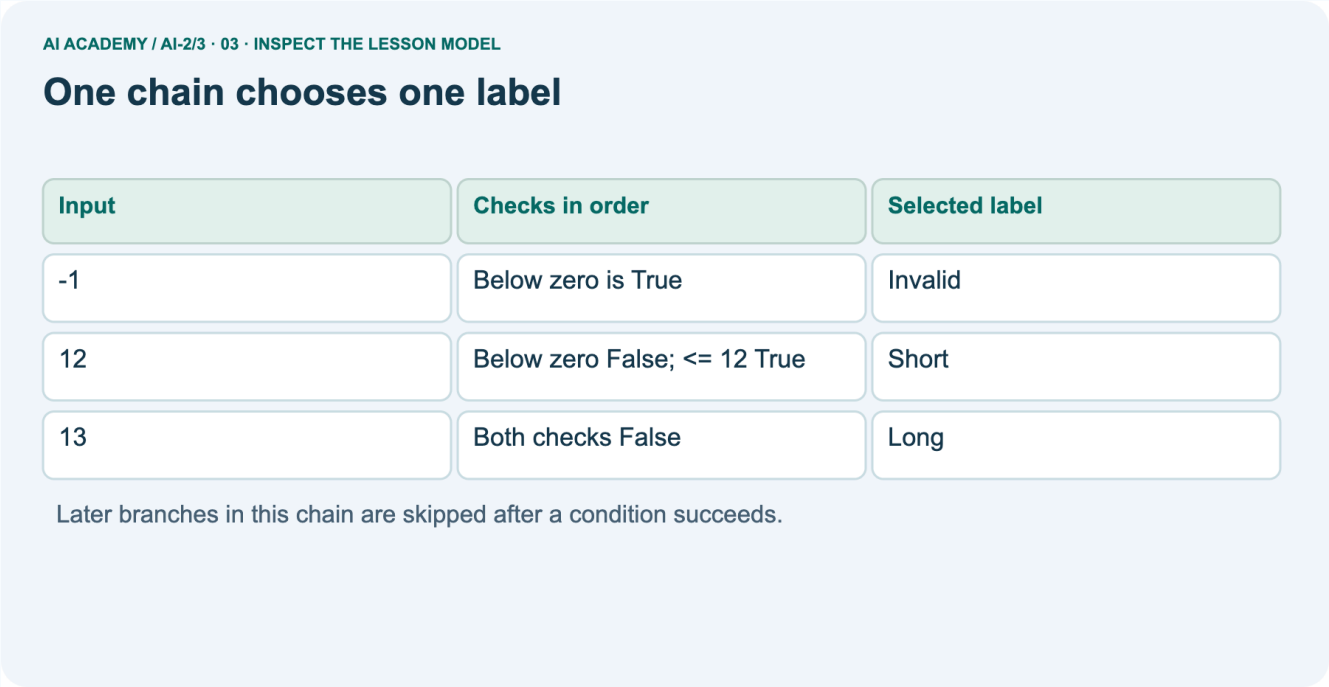
Follow the labelled picture with the matching explanation. Open the vector diagram

Negative check: −1 takes Invalid and skips later **branch**es. → Zero check: 0 takes Empty, not Short. → Inclusive short range: 1 and 10 take Short. → Remaining **input**s: 11 takes Long; only the selected **branch** assigns the label.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. One chain chooses one label



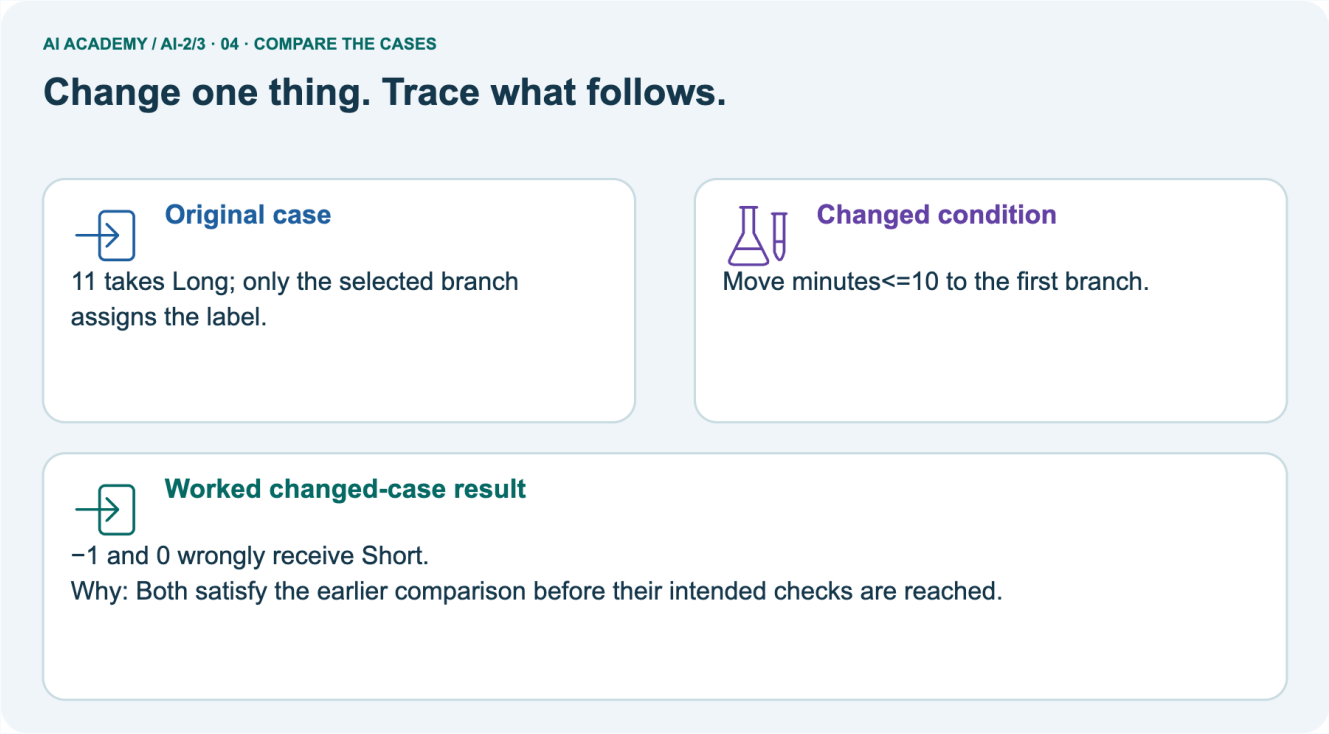
Follow the labelled picture with the matching explanation. Open the vector diagram

One chain chooses one label. -1 | Below zero is True | Invalid. 12 | Below zero False; <= 12 True | Short. 13 | Both checks False | Long

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: 11 takes Long; only the selected **branch** assigns the label. Changed **condition**: Move minutes<=10 to the first **branch**. Expected result: −1 and 0 wrongly receive Short.

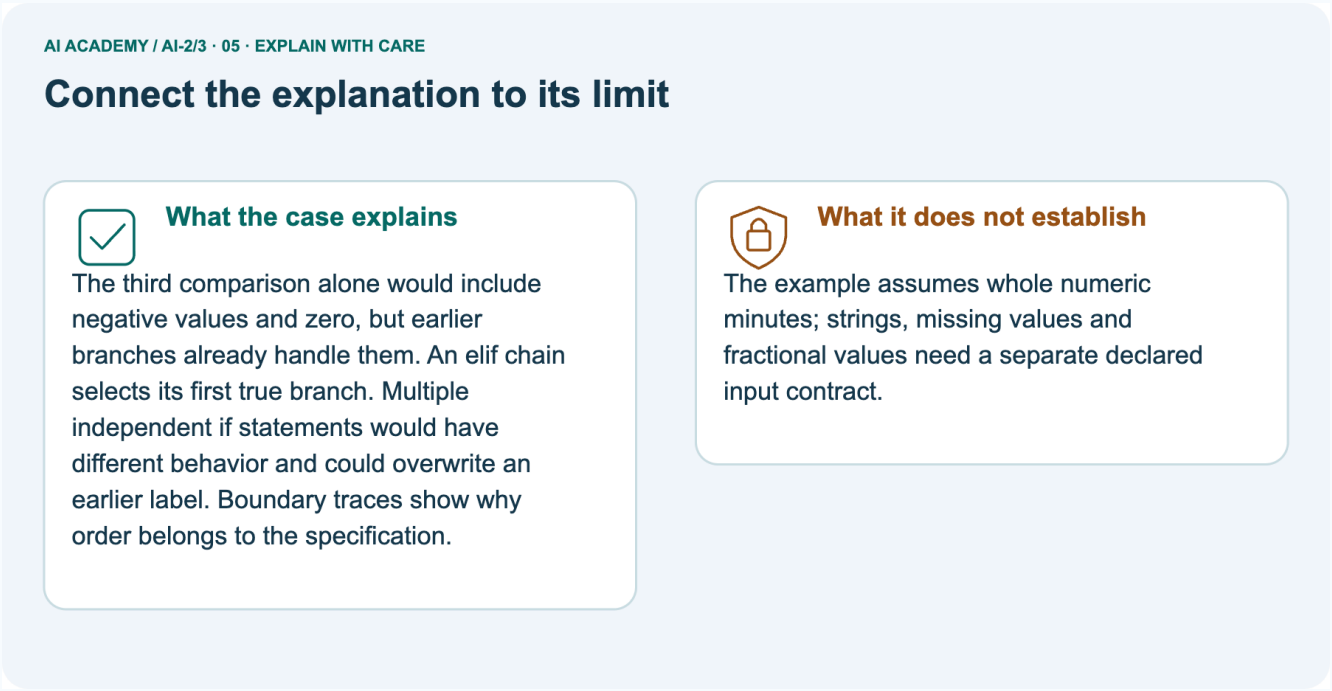
Reason: Both satisfy the earlier comparison before their intended checks are reached.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Both satisfy the earlier comparison before their intended checks are reached.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The third comparison alone would include negative values and zero, but earlier **branch**es already handle them. An elif chain selects its first true **branch**.

Multiple independent if statements would have different behavior and could overwrite an earlier label. **Boundary** traces show why **order** belongs to the specification.

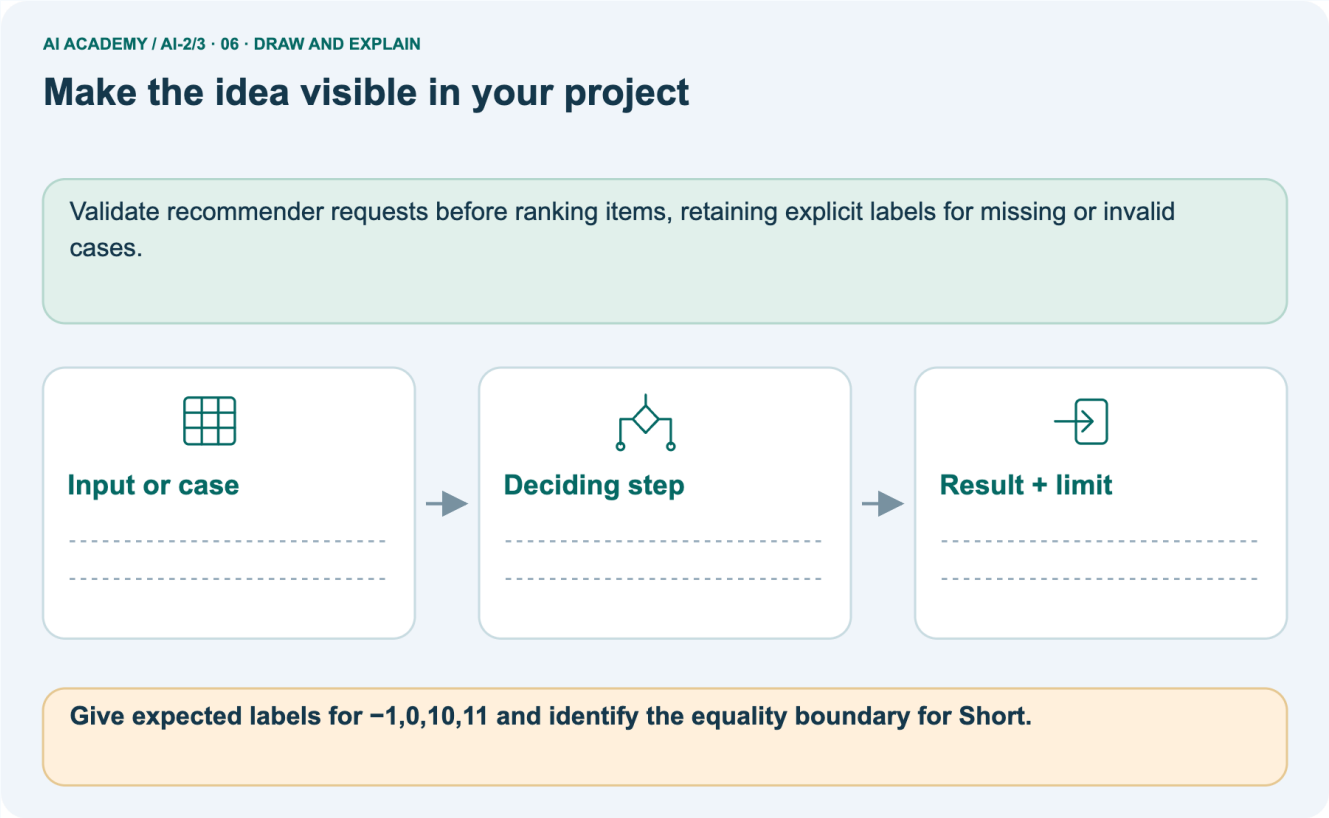
Limit: The example assumes whole numeric minutes; strings, missing values and fractional values need a separate declared **input** contract.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: The example assumes whole numeric minutes; strings, missing values and fractional values need a separate declared **input** contract.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Validate recommender requests before ranking items, retaining explicit labels for missing or invalid cases. Draw your **input**, deciding step, result and limit.

Supported question: Give expected labels for −1,0,10,11 and identify the equality **boundary** for Short.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Invalid, Empty, Short, Long. Ten is included in Short because the comparison uses <=.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Branching Logic: the case

02. Follow the deciding steps

03. One chain chooses one label

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.